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ContextDrive: A Smartphone Based Context-Aware Driver Assistance System

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- ContextDrive is a smartphone based, context-aware driver assistance system designed to improve road safety by assessing driving risk in real time.
- The system integrates camera input, GPS data, weather information, and time based context to analyze multiple driving factors together rather than considering individual hazards independently.
- Based on the combined context, the system estimates the current driving risk and provides explainable safety recommendations to the driver.
- The proposed solution offers an affordable, portable, and practical approach to enhancing driver awareness without requiring additional hardware.

- Existing driver assistance systems use technologies such as cameras, GPS, and other sensors to detect individual hazards and assist drivers during driving.
- However, many systems focus on specific hazards rather than combining multiple factors such as vehicle surroundings, driving speed, weather, and visibility to understand the overall driving context.
- **ContextDrive** is a smartphone based context-aware driver assistance system that integrates these factors to continuously assess driving risk and provide explainable safety recommendations in real time.
- The proposed system aims to provide:
 - **Context-Aware Risk Assessment**
 - **Explainable Recommendations**
 - **Affordable Solution**

Existing affordable, smartphone based driver assistance solutions evaluate hazards like speed, weather, visibility, and traffic in isolation rather than modeling how these factors compound and interact, creating a need for a smartphone only system that estimates real time driving risk from these contextual interactions and provides explainable safety recommendations.

- Can be used in any vehicle by simply mounting a smartphone, eliminating the need for expensive additional hardware.
- Assists drivers by analyzing real time driving conditions using camera, GPS, weather, and time based information.
- Provides context-aware risk assessment and explainable safety recommendations to improve driver awareness and promote safer driving.

OBJECTIVES

- To develop a smartphone based context-aware driver assistance system for real time driving risk assessment.
- To integrate camera, GPS, weather, and time based information for comprehensive driving context analysis.
- To generate an explainable driving risk score and provide timely safety recommendations.
- To improve driver awareness and promote safer driving without requiring additional hardware.

Personalised Driver Risk Assessment with Adaptive Feedback using Crowdsensed Telemetric Data

- **Authors:** Auwal Sagir Muhammad et al.
- **Published in:** IET Intelligent Transport Systems (2025), IET
- **Methodology:** Proposes GPS derived driver risk indicators, including thresholds for harsh braking, rapid acceleration, overspeeding, turning rate computed from bearing changes, and contextual flags based on night time and rainy conditions. These engineered indicators provide a structured approach for extracting interpretable driving risk features from telemetric data.
- **Advantages:** Introduces validated feature engineering methods for transforming raw GPS, temporal, and weather information into meaningful driver behaviour indicators using well defined threshold based rules.
- **Disadvantages:** The proposed thresholds were validated within a machine learning pipeline, therefore, their numerical values may require recalibration when applied to different risk assessment frameworks.

A New CNN Based Object Detection System for Autonomous Mobile Robots Based on Real World Vehicle Datasets

- **Authors:** Udink Aulia et al.
- **Published in:** Heliyon (2024), Elsevier
- **Methodology:** Develops a lightweight vehicle detection system using the SSD MobileNetV2 FPN-Lite architecture with transfer learning. The pretrained detector is fine tuned on a real world vehicle dataset collected and annotated by the authors to improve detection performance in the target deployment environment.
- **Advantages:** Demonstrates that transfer learning and fine tuning of a lightweight SSD MobileNetV2 detector can significantly improve vehicle detection accuracy while maintaining real time performance, providing a practical approach for domain specific deployment.
- **Disadvantages:** Detection performance depends on the quality and diversity of the fine tuning dataset, requiring additional data collection and annotation when adapting the model to new environments.

Using Contextual Data to Predict Risky Driving Events: A Novel Methodology from Explainable Artificial Intelligence

- **Authors:** Leandro Masello et al.
- **Published in:** Accident Analysis & Prevention (2023), Elsevier
- **Methodology:** Investigates the influence of contextual driving factors, including speed limit, weather, traffic conditions, road characteristics, and environmental variables, on risky driving events.
- **Advantages:** Demonstrates that driving risk is determined by the combined influence of multiple contextual factors rather than isolated variables, providing an evidence based foundation for context-aware risk assessment and feature weighting.
- **Disadvantages:** The proposed methodology relies on supervised machine learning and large scale telematics data, requiring retraining and recalibration when applied to different driving environments or datasets.

Optimizing Monocular Driving Assistance for Real Time Processing on Jetson AGX Xavier

- **Authors:** Huy-Hung Nguyen et al.
- **Published in:** IEEE Access (2024), IEEE
- **Methodology:** Presents a real time monocular camera based driver assistance system that employs lightweight object detection and vision based perception to identify road users and support multiple ADAS functions on an edge device.
- **Advantages:** Demonstrates that a single monocular camera, combined with lightweight object detection, can reliably perform real time perception tasks for driver assistance, validating the feasibility of camera based road scene understanding without specialized sensors.
- **Disadvantages:** The system is optimized for Jetson AGX Xavier hardware, making its deployment strategy and performance improvements less directly transferable to other edge platforms and smartphone based implementations.

Balancing Transparency and Control: The Impact of AI Explanation Detail on User Perception in Automated Vehicles

- **Authors:** Jakob Peintner et al.
- **Published in:** Transportation Research Interdisciplinary Perspectives (2025), Elsevier
- **Methodology:** Investigates the effect of different AI explanation levels (None, “How”, “How & Why”, and “Contrastive & Counterfactual”) on user perception through a virtual reality driving study, evaluating their influence on trust, user experience, and perceived safety.
- **Advantages:** Demonstrates that concise “How & Why” explanations provide an effective balance between transparency and usability, offering an evidence based approach for designing understandable driver assistance recommendations.
- **Disadvantages:** The findings are based on passenger interactions in a simulated Level-4 automated driving environment and may not directly generalize to real time driver assistance systems where users actively control the vehicle.

LITERATURE SURVEY SUMMARY

Paper	Methodology	Advantages	Disadvantages
Personalised Driver Risk Assessment with Adaptive Feedback	GPS derived driver risk indicators using speed, harsh braking, acceleration, turning rate, and contextual driving information.	Provides validated feature engineering methods for transforming raw GPS, temporal, and weather data into interpretable driver behaviour indicators.	Threshold values were validated within a machine learning pipeline and may require recalibration for different risk assessment frameworks.
A New CNN-Based Object Detection System for Autonomous Mobile Robots	SSD MobileNetV2 FPN-Lite with transfer learning and fine-tuning on a real world vehicle dataset.	Demonstrates that transfer learning and fine-tuning improve lightweight vehicle detection while maintaining real time performance.	Performance depends on the quality and diversity of the fine-tuning dataset, requiring additional data collection and annotation.
Using Contextual Data to Predict Risky Driving Events	Analyse the influence of contextual driving factors such as speed limit, weather, traffic, and road characteristics on risky driving events.	Shows that driving risk depends on the combined influence of multiple contextual factors, supporting context-aware risk assessment and feature weighting.	Relies on supervised machine learning and large scale telematics data, requiring retraining for different deployment environments.
Optimizing Monocular Driving Assistance for Real Time Processing	Real time monocular camera based driver assistance using lightweight object detection and vision based perception.	Validates that a single monocular camera can support reliable real time driver assistance without specialized sensors.	Optimized specifically for Jetson AGX Xavier hardware, limiting direct applicability to smartphone platforms.
Balancing Transparency and Control: The Impact of AI Explanation Detail	Evaluates different AI explanation levels through a virtual reality user study to measure trust, usability, and perceived safety.	Shows that concise "How & Why" explanations provide the best balance between transparency and usability.	Findings are based on a simulated Level-4 automated driving environment and may not directly generalize to driver assistance systems.

Hardware Requirements

- **Smartphone** : Serves as the target platform, providing processing capability and internet connectivity.
- **Rear Camera**: Captures live road scenes for vehicle and vulnerable road user detection.
- **GPS Receiver**: Provides real time speed, location, and bearing information for driving behaviour analysis.
- **Internet Connectivity**: Retrieves real-time weather information through the OpenWeatherMap API.

Software Requirements

- **Programming Language:** Dart 3.12
- **Application Framework:** Flutter 3.44
- **Object Detection Framework:** TensorFlow Lite 2.20
- **Object Detection Model:** SSD MobileNetV2 FPN-Lite
- **Weather Service:** OpenWeatherMap API

- The proposed system is a smartphone based context-aware driver assistance system that continuously monitors the driving environment and estimates driving risk in real time.

Key Features of the Design:

- **Context-Aware Analysis**
- **Real Time Risk Assessment**
- **Explainable Recommendations**
- **Smartphone Based Solution**

Technical Feasibility

- **Technology Stack:** Developed as a mobile application using Flutter, with TensorFlow Lite for real time object detection.
- **Hardware:** Requires only a smartphone equipped with a rear camera, GPS receiver, and internet connectivity, eliminating the need for specialized sensors or external hardware.
- **Real Time Processing:** Modern smartphones provide sufficient computational capability to perform on-device object detection, contextual risk assessment, and explainable safety recommendations in real time.

Economic Feasibility

- **Development Cost:** Requires only a smartphone, freely available development tools, and a low cost weather API service.
- **Benefits:**
 - Provides a cost effective smartphone based driver assistance solution without additional hardware.
 - Enhances driver awareness and promotes safer driving, reducing the risk of road accidents.

ARCHITECTURE DIAGRAM

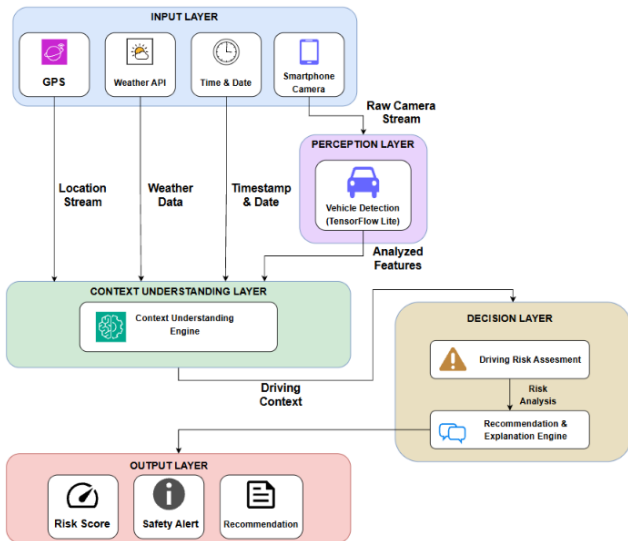


Figure 1: System Architecture Diagram

USE CASE DIAGRAM

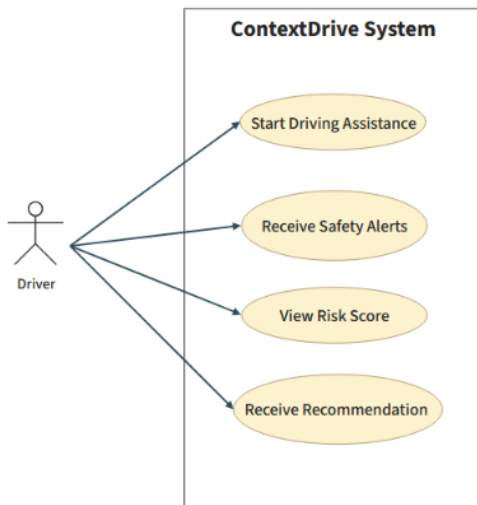


Figure 2: Use Case Diagram



Figure 3: Data Flow Diagram - Level 0

DATA FLOW DIAGRAM - LEVEL 1

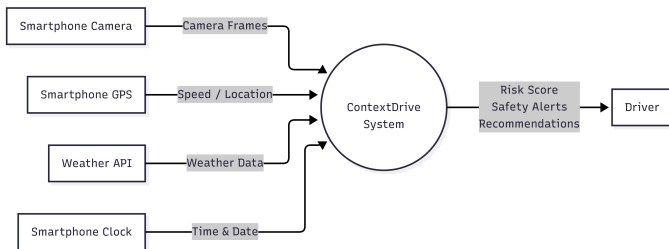


Figure 4: Data Flow Diagram - Level 1

DATA FLOW DIAGRAM - LEVEL 2

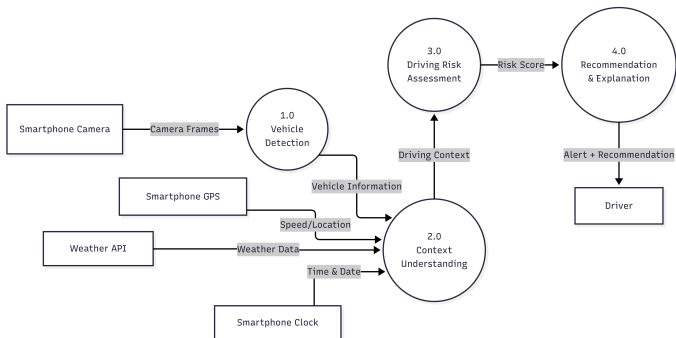


Figure 5: Data Flow Diagram - Level 2

GANTT CHART

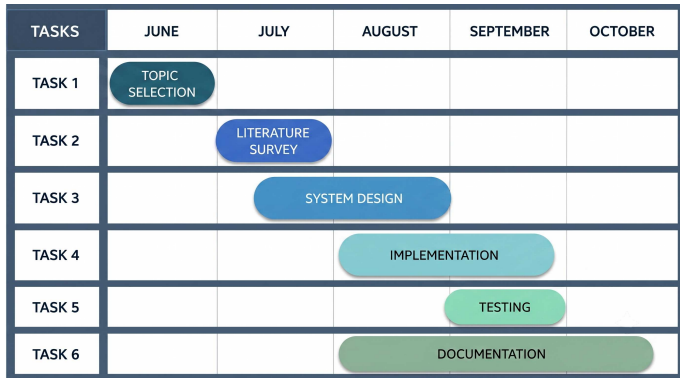


Figure 6: Gantt Chart

- The proposed **ContextDrive** system aims to provide a smartphone based, context-aware driving assistance solution using real time data from the camera, GPS, weather, and time.
- The system integrates multiple contextual factors to assess driving risk and generate intelligent safety recommendations.
- The proposed architecture is designed to perform on device processing, making it practical, portable, and suitable for everyday use.
- The project is expected to improve driver awareness and promote safer driving through timely risk assessment and explainable recommendations.

- [1] A. S. Muhammad, L. Chen, and C. Wang, "Personalised driver risk assessment with adaptive feedback using crowdsensed telemetric data," *IET Intelligent Transport Systems*, vol. 19, no. 1, p. e70071, 2025.
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- [4] H.-H. Nguyen, D. N.-N. Tran, L. H. Pham, and J. W. Jeon, "Optimizing monocular driving assistance for real-time processing on jetson agx xavier," *IEEE Access*, vol. 12, pp. 71 853–71 865, 2024.
- [5] J. Peintner, C. Manger, M. Berger, and A. Riener, "Balancing transparency and control: The impact of ai explanation detail on user perception in automated vehicles," *Transportation Research Interdisciplinary Perspectives*, vol. 34, p. 101742, 2025.

THANK YOU